CO2-AT1

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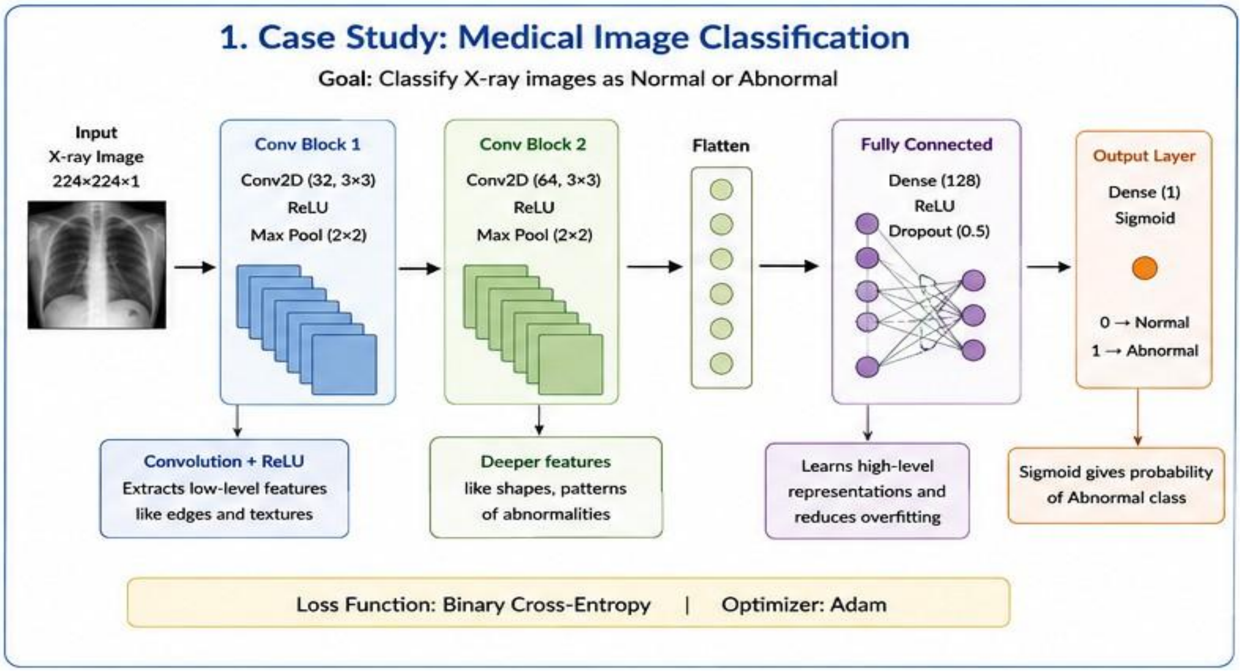
COURSE CODE:MLA0401
COURSE NAME:DEEP LEARNING

Question 1. Medical Image Classification



Explanation

1. Input Layer:
The X-ray is resized to **224 × 224 pixels**. Since an X-ray is grayscale, it has one channel.

2. Convolution Layer:
The first convolution layer uses filters to detect simple features such as **edges, lines and textures**.

3. ReLU:
ReLU adds non-linearity and helps the network learn complex patterns.

ReLU(x)=max(0,x)

4. Max Pooling:
Pooling reduces the size of feature maps while retaining important features. This reduces computation.

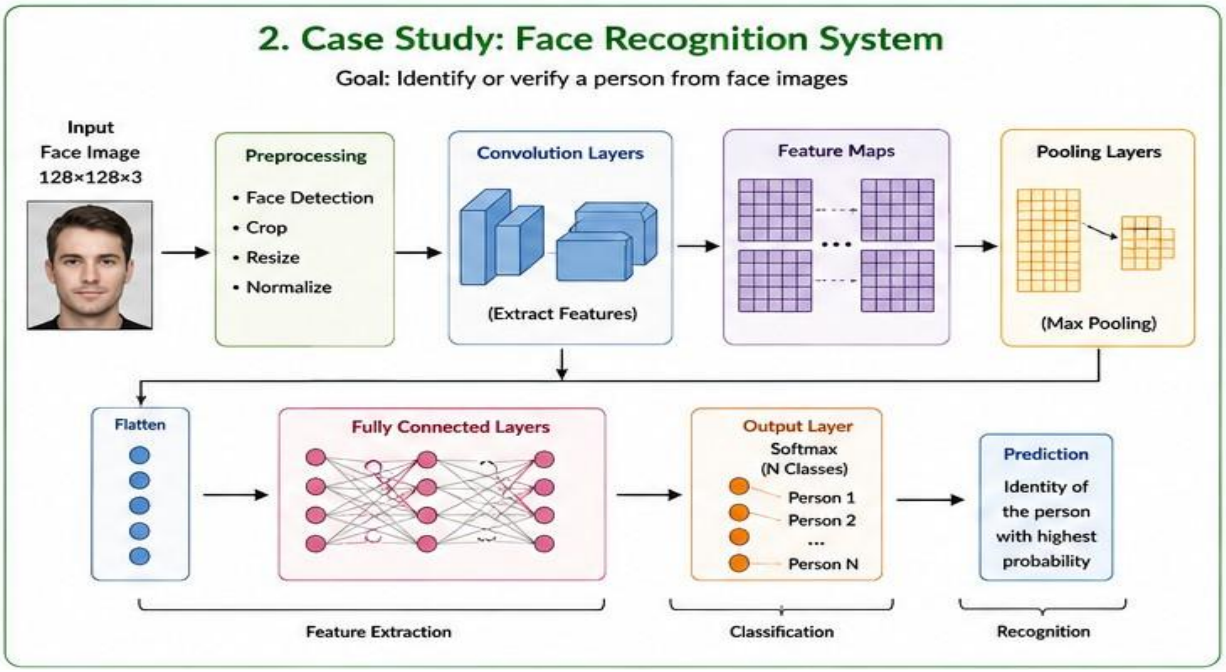
5. Flatten:
The feature maps are converted into a one-dimensional vector.

6. Fully Connected Layer:
The dense layer combines the extracted features and learns high-level representations.

9. Output Layer:
Since there are two classes, a **Sigmoid** output is suitable.

- Output close to 0 → Normal
- Output close to 1 → Abnormal

Question 2. Face Recognition System



Explanation

1. Face Image:
The camera captures an image containing a person's face.

2. Preprocessing:
The face is detected, cropped and resized. Normalization makes the image suitable for CNN processing.

3. Convolution:
CNN filters detect facial features.

Early layers detect:

- Edges
- Lines
- Curves

Deeper layers detect:

- Eyes
- Nose
- Mouth
- Face shape

4. Feature Maps:
Each filter creates a feature map showing where a particular feature is present.

5. Pooling:
Max pooling reduces the feature-map dimensions and keeps important information.

6. Fully Connected Layer:
The extracted features are combined to create a representation of the face.

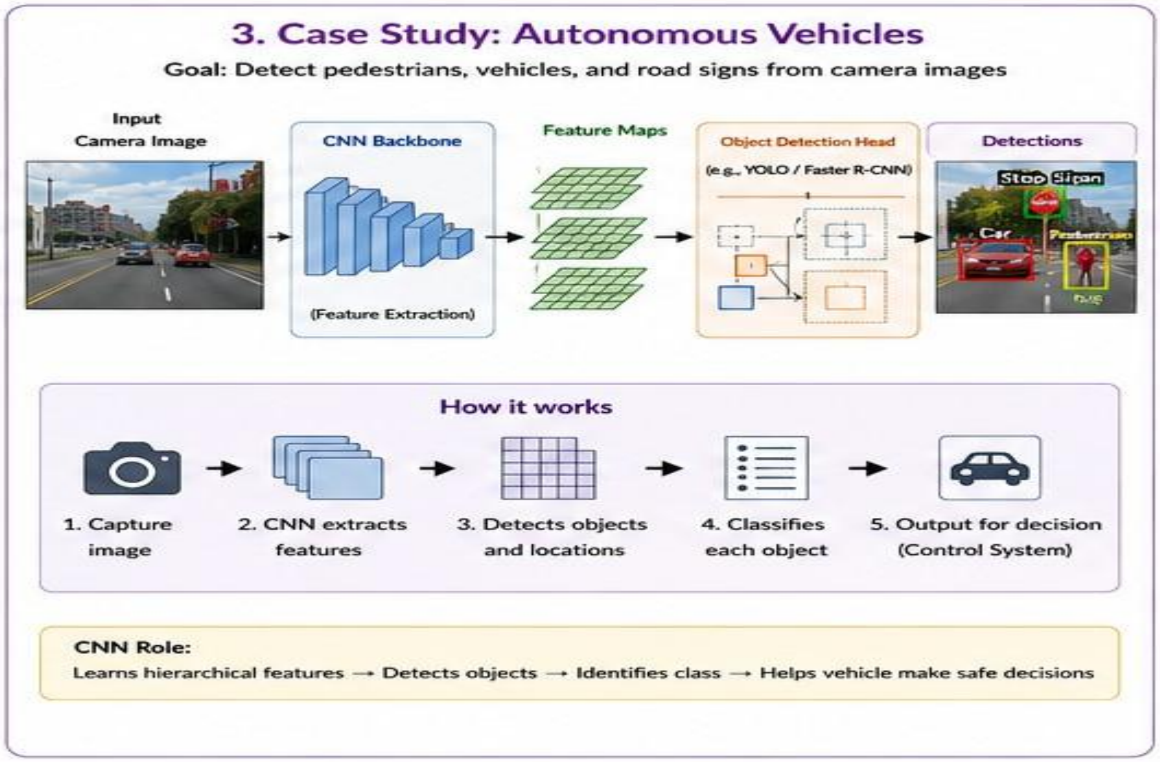
7. Classification:
The final layer can compare the representation with registered student identities.

For example:

Student A → 10%
Student B → 85%
Student C → 5%

The system identifies **Student B**.

Question 3.Autonomous Vehicles



Explanation

1. Camera:
The vehicle continuously captures road images.

2. CNN Backbone:
The CNN processes the image and extracts important visual features.

3. Feature Extraction:
Early layers detect edges and shapes. Deeper layers detect objects such as cars, humans and signs.

4. Feature Maps:
The extracted features are represented using feature maps.

5. Object Detection:
Object detection identifies **both the object and its location**.

For example:

Car → 95% confidence
Pedestrian → 92% confidence
Stop Sign → 98% confidence

A bounding box is drawn around each detected object.

6. Decision Making:

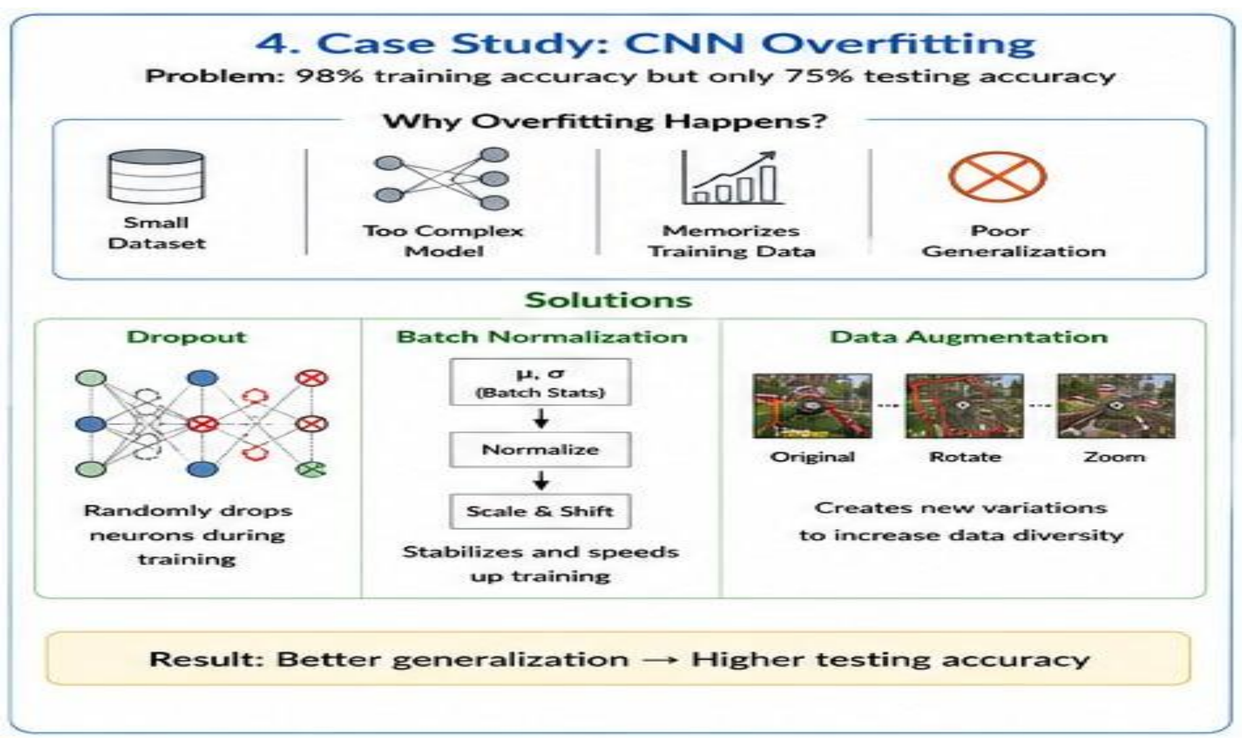
If a pedestrian is detected:

Pedestrian detected
↓
Vehicle slows down
↓
If necessary → Stop

If a stop sign is detected:

Stop Sign
↓
Vehicle Control
↓
Stop

Question 4. CNN Overfitting



Why Does Overfitting Occur?

1. Small Dataset

If there are not enough training images, the CNN may memorize the training examples instead of learning general patterns.

2. Complex Model

A CNN with too many layers, filters or neurons can memorize training data.

3. Lack of Data Diversity

If all training images look similar, the model may perform poorly on different images.

4. Excessive Training

Training for too many epochs can cause the model to memorize the training data.

Solutions

A. Dropout

Input
↓
Neurons
↓
Some neurons randomly removed
↓
Output

For example, **Dropout = 0.5** means approximately half of the selected neurons are randomly dropped during each training step.

Benefits:

- Reduces overfitting.
- Prevents neuron dependency.
- Improves generalization

B. Batch Normalization

Input
↓

Convolution
↓
Batch Normalization
↓
ReLU
↓
Next Layer

Batch normalization normalizes intermediate activations.

Benefits:

- Stabilizes training.
- Often speeds up convergence.
- Can provide some regularization.

C. Data Augmentation

One training image can be modified into several variations.

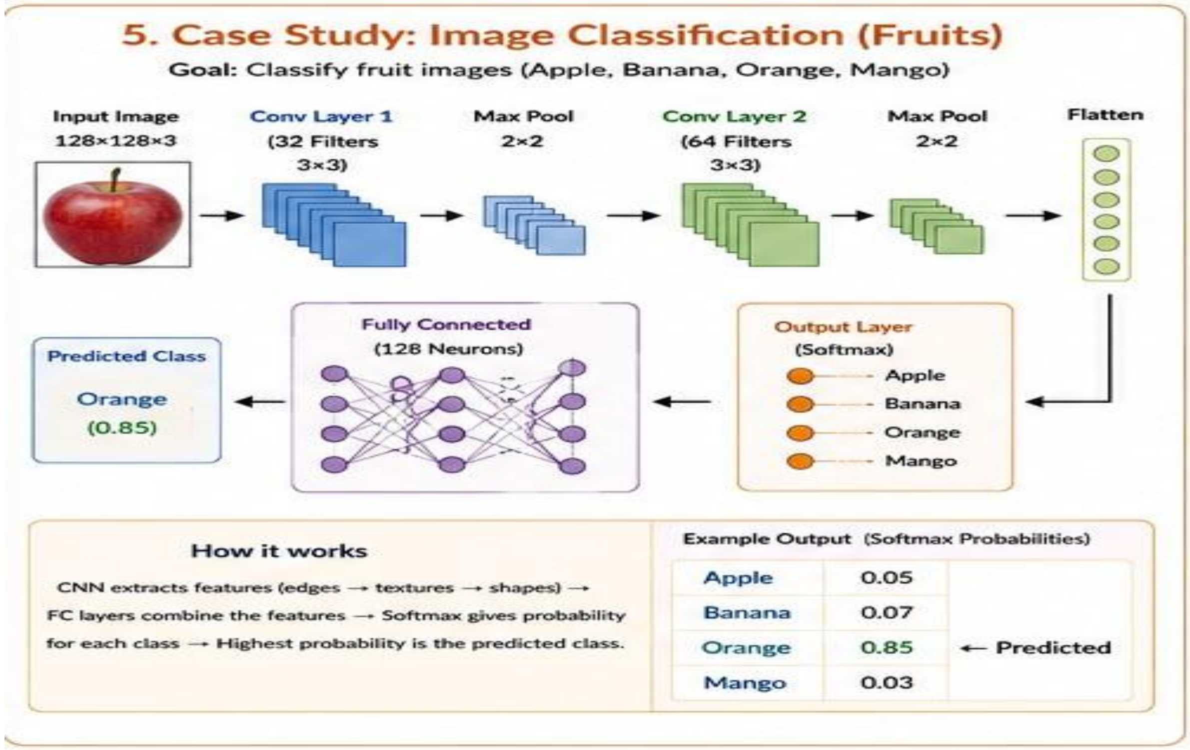
Original Image
↓

↓ ↓ ↓ ↓
Rotate Flip Zoom Crop
↓ ↓ ↓ ↓
New Training Images

Benefits:

- Increases data diversity.
- Reduces overfitting.
- Improves model robustness.

Question 5. Fruit Image Classification



Explanation

1. Input Image:
The RGB fruit image is resized to **128 × 128 × 3**.

The 3 represents:

- Red
- Green
- Blue

2. First Convolution:
The CNN detects simple features such as:

- Edges
- Colors
- Curves
- Basic textures

3. ReLU:
ReLU introduces non-linearity and allows the network to learn complex patterns.

4. Max Pooling:
Reduces the size of the feature maps while preserving important features.

5. Second Convolution:
This layer learns higher-level features such as:

- Fruit shape
- Texture
- Surface patterns
- Color combinations

6. Flatten:
Converts the feature maps into a one-dimensional vector.

7. Fully Connected Layer:
Combines all extracted features to determine the type of fruit.

8. Softmax Output:
There are four classes, so the output layer has **4 neurons**.: